

Social Networking & Graph Algorithms

Omar Marey (900245561), Ahmed Yamany (900235125), Tarek Kamel (900233638), Adham Abotarboush (900243583)
Department of Computer Science, The American University in Cairo (AUC)

Abstract—Social networks can be modeled as graphs where nodes represent users (or entities) and edges represent interactions. This paper surveys core graph algorithms used to analyze social networks, focusing on connectivity, influence, and community structure. We review breadth-first search (BFS) and depth-first search (DFS) for reachability and component analysis; shortest-path methods for information diffusion and distance metrics; PageRank-style centrality for influence ranking; and fast, scalable community detection (e.g., label propagation and modularity-based approaches) for uncovering group structure. We discuss how data structures (adjacency lists and compressed sparse row formats) affect runtime and memory on large, sparse social graphs, and outline trade-offs for dynamic graphs where edges change over time. Building on the survey, we propose a small experiment: compare two community detection algorithms (Label Propagation vs. Louvain-style modularity optimization) on a publicly available social subgraph, evaluating runtime, modularity, and stability across runs. Results will illustrate how algorithmic and data-structure choices impact performance and interpretability in practical student-scale settings. The paper concludes with lessons for choosing algorithms under resource constraints and directions for further study (dynamic graphs, weighted and directed networks).

I. INITIAL PLAN

• Introduction

- Motivate why social graphs matter: reachability, influence, communities.
- State paper goals, contributions, and organization.

• Background: Graph Models & Data Structures

- Simple vs. directed/weighted graphs; sparsity in social networks.
- Data structures for large graphs: adjacency lists vs. CSR; memory/runtime implications.
- Brief note on dynamic graphs (edge churn and updates).

• Core Algorithms for Social Networks

- Traversals: BFS/DFS for components and diameter approximations.
- Shortest paths: unweighted (BFS) vs. weighted (Dijkstra at a glance); diffusion and distance use cases.
- Centrality (intuition-first): degree, betweenness (high-level), PageRank-style influence.
- Community detection: Label Propagation (LPA) and modularity-based (Louvain-style); pros/cons and when to use which.

• Experiment Design (sketch for next milestone)

- Task: Compare LPA vs. Louvain-style on the same social subgraph.

• Results & Discussion

- Tables/plots of metrics; qualitative interpretation; limitations (randomness, small graphs).
- Overall reflection upon how the research paper educated and aided us on the term project.

• Conclusion & Future Work

- Practical guidelines for student projects; pointers to weighted/directed and dynamic graphs.

• References (to be populated)